Discovery-agent → structural-modeling handoff

Per-candidate record in structural_modeling_handoff.json — three blocks are deterministically derived by the agent (green)

candidate record

identity

accession	*E2JA32*
protein_name	*... diaminopropanoate ligase*
organism, gene_names, ec_numbers	

family — NEW

interpro_family	*Dapdiamide ligase*
interpro_family_id	*IPR048223*

evidence (existing)

- structure_sources
- native_substrate_evidence
- reaction_chemistry
- structural_homology / orthologs

mechanism_evidence.activation — NEW

activation_mode	*acyl_adenylate*
reactive_phosphate	*Palpha*
nucleophile	*carboxylate O*
confidence	*high*
source / derived_from	*reaction products*

modeling_input — NEW

cofactors[] {name,ccd,count,role,basis}	*ATP×1 + MG×2*
reactive_phosphate	*Palpha*
activation_mode	*acyl_adenylate*

how the green blocks are derived (worked example: ddaG)

1 · candidate's own annotations (UniProt)

native reaction:

...+ ATP = ...+ AMP + diphosphate

cofactor comment: (none for ddaG)

deterministic (no LLM)

2 · chemistry_logic derivations

derive_activation_mechanism()

AMP + diphosphate → α-phosphorus

derive_modeling_cofactors()

ATP×1; metal MG×2 (mech. default)

3 · written to handoff (with provenance)

reactive_phosphate: Palpha

activation_mode: acyl_adenylate

cofactors: ATP×1, MG×2

basis: reaction products / uniprot

confidence: high

Why it matters

Family name 'Dapdiamide ligase' gives no hint of adenylation. The reaction pins it to Pα — fixing the old family-keyword mis-call that scored ddaG on Pγ.

existing field — agent-derived (deterministic)